

# Theoretical Foundations of Option Pricing Models

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## 1 Core Assumptions of the Black-Scholes World

The foundational models described here operate under a specific set of idealized market conditions known as the "Black-Scholes world." These assumptions are crucial for the derivation of the formulas.

- **Random Walk:** The stock price follows a Geometric Brownian Motion, meaning price changes are random and unpredictable.
- **Constant Volatility and Interest Rates:** The volatility ( $\sigma$ ) of the underlying asset and the risk-free interest rate ( $r$ ) are known and constant over the life of the option.
- **No Dividends:** The original model assumes the underlying stock does not pay dividends. Extensions (like Merton's model) adjust for this.
- **Frictionless Market:** There are no transaction costs (e.g., commissions or brokerage fees) or taxes. Securities are perfectly divisible.
- **Efficient Markets:** All relevant information is immediately reflected in the stock price. Arbitrage opportunities do not exist.
- **Lognormal Distribution:** Stock prices are lognormally distributed, which means that the logarithm of the stock price is normally distributed.

## 2 Barone-Adesi and Whaley (1987) Model for American Options

American options can be exercised at any time up to expiration, a feature that complicates pricing. As there is no simple closed-form solution, models like Barone-Adesi and Whaley provide a highly accurate and fast quadratic approximation.

### 2.1 Concept

The value of an American option is the sum of its European equivalent price and a premium for the early exercise feature.

$$V_{\text{American}} = V_{\text{European}} + \text{Early Exercise Premium}$$

The model works by finding a critical stock price ( $S^*$ ), explained further in Section 3, below (for puts) or above (for calls) which it is optimal to exercise the option early.

### 2.2 Formulas

For an American call option, if  $S < S^*$ , the price is:

$$C(S) \approx C_{BS}(S) + A_2 \left( \frac{S}{S^*} \right)^{q_2}$$

If  $S \geq S^*$ , the price is simply  $S - K$ .

For an American put option, if  $S > S^*$ , the price is:

$$P(S) \approx P_{BS}(S) + A_1 \left( \frac{S}{S^*} \right)^{q_1}$$

If  $S \leq S^*$ , the price is  $K - S$ .

The terms  $q_1$ ,  $q_2$ ,  $A_1$ , and  $A_2$  are derived from the underlying Black-Scholes framework. The critical price  $S^*$  is found iteratively using a numerical solver like the Newton-Raphson method.

### 2.3 Parameters Used

| Parameter               | Value      |
|-------------------------|------------|
| S (Current Stock Price) | 100        |
| K (Strike Price)        | 100        |
| T (Time to Maturity)    | 1.0 year   |
| r (Risk-free Rate)      | 0.05 (5%)  |
| $\sigma$ (Volatility)   | 0.20 (20%) |
| q (Dividend Yield)      | 0.03 (3%)  |

### 2.4 Numerical Results

| Option Type   | Calculated Price |
|---------------|------------------|
| American Call | \$9.4335         |
| American Put  | \$6.4343         |

Note: The calculated price is the premium per share for one American option contract. This represents the cost (\$9.4335 per option contract for the call) to acquire the right to exercise the option at any point until its expiration.

## 2.5 Limitations

While very accurate for a wide range of parameters, it is still an approximation. Its accuracy can decrease for options with very long maturities or unusual parameter combinations when compared to more computationally intensive methods.

## 3 Calculating the Critical Stock Price ( $S^*$ )

The critical stock price,  $S^*$ , is a pivotal concept in the pricing of American options. It represents the point of indifference for an option holder. At this specific stock price, the investor is perfectly torn between two choices:

1. Exercising the option immediately to receive its intrinsic value (either  $S^* - K$  for a call or  $K - S^*$  for a put).
2. Holding the option longer to retain its remaining time value and the possibility of greater future profits.

At the critical price  $S^*$ , the value derived from these two actions is identical. The goal is therefore to find the value of  $S^*$  that solves the equation where the option's model price is exactly equal to its exercise value.

### 3.1 The Newton-Raphson Method

Because the option pricing formula is complex, this equation cannot be solved with simple algebra. Instead, a numerical root-finding algorithm called the Newton-Raphson method is used. This method works iteratively:

- **Initial Guess:** The algorithm starts with a reasonable first guess for  $S^*$ .
- **Calculate and Check:** It plugs this guess into the equation and checks how far the result is from zero.
- **Refine the Guess:** Using calculus (specifically, the derivative of the function), the method determines the direction and magnitude of the error. It uses this information to make a new, much more accurate guess.
- **Repeat:** The process repeats, with each new guess getting progressively closer to the true solution. After a few iterations, the guess is so precise that the equation is solved, and that final guess is determined to be the critical price,  $S^*$ .